BUSFIN 8200 – Problem Set 2

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Question 1

Question 2

2 (a)

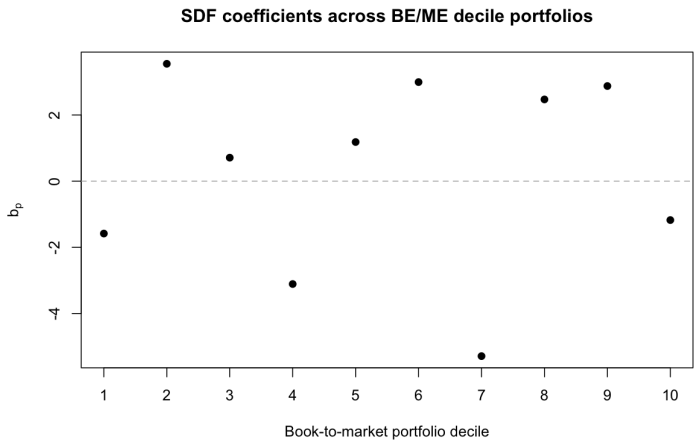
Using the normalization $\mathbb{E}[M^*] = 1$, we find the stochastic discount factor $M_t^* = a - b'xR_t$ that satisfies $\hat{\mathbb{E}}[M^* \cdot xR] = 0$. In particular, we have:

$$\begin{aligned} 0 &= \mathbb{E}[M^* \cdot xR] \\ &= \underbrace{\mathbb{E}[M^*]}_{=1} \cdot \mathbb{E}[xR] + \text{Cov}(M^*, xR) \\ &= \mathbb{E}[xR] - b'\text{Cov}(xR, xR) \\ &\implies \mathbb{E}[xR] = b'\Sigma \\ &\implies b = \Sigma^{-1}\mathbb{E}[xR] \end{aligned}$$

xR is a Tx10 vector of excess returns with T = 1,202 months in the sample from July 1926 to August 2026. Verifying that Σ is invertible, we find that:

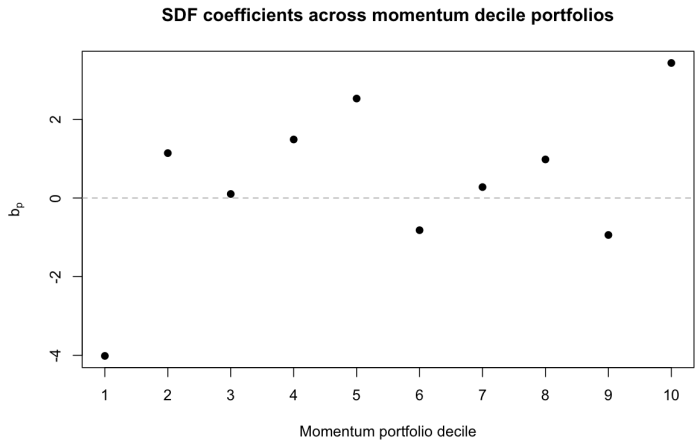
$$\begin{aligned} 1 &= \mathbb{E}[M^*] \\ &= \mathbb{E}[a - b'xR] \\ &= a - b'\mathbb{E}[xR] \\ &\implies a = 1 + b'\mathbb{E}[xR] \end{aligned}$$

Therefore, we are able to compute the stochastic discount factor $M_t^* = a - b'xR_t$ for each month in the sample and obtain the following scatterplot:



2 (b)

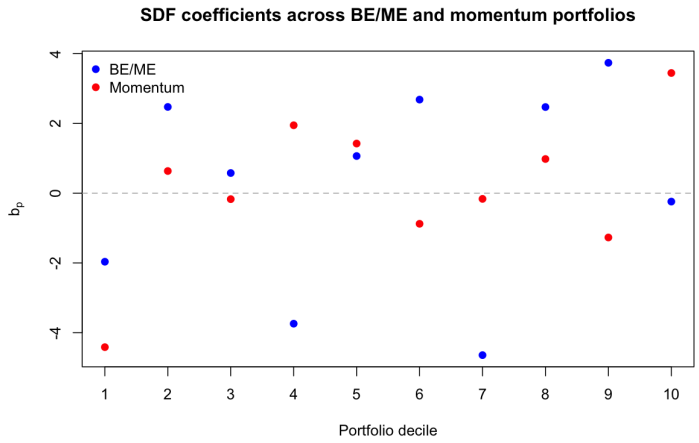
Following the same procedure as in part (a), we find the respective b coefficients of the stochastic discount factor that prices the excess returns of the 10 portfolios in the MOM dataset as follows:



Similar to part (a), we do not observe any clear pattern between the b coefficients and the decile portfolios' excess returns. The correlation between the recovered M_t^* is 0.3555, which suggests that the SDF that prices a particular portfolio does not necessarily price the other one as the decile portfolios are sorted using different signals.

2 (c)

Following the same procedure as in part (a), we find the respective b coefficients of the stochastic discount factor that prices the excess returns of the 20 portfolios in the BM & MOM datasets as follows:



Using a common sample of 1,196 months from July 1927 to August 2026, we estimated the Sharpe ratios of xR_t^* as follows:

Table 1: Monthly Sharpe Ratios of the xR^* Portfolios

Portfolio set	Monthly Sharpe ratio
BE/ME (10 portfolios)	0.191841
Momentum (10 portfolios)	0.257990
Combined (20 portfolios)	0.304398

A key property of the SDF under no arbitrage is that xR_t^* is the maximum Sharpe ratio portfolio. Since we observed that the Sharpe ratio of the combined 20 portfolios is higher than the Sharpe ratios of the individual 10 portfolios, adding more non-redundant assets to the portfolio universe provides a more diversified portfolio with better risk-adjusted returns and higher volatility, which is consistent with the theory of SDF and maximum Sharpe ratio portfolios.

2 (d)

Having estimated the SDFs $M_t^* = \hat{a} - \hat{b}' xR_t$ from part (a),(b), and (c) and normalizing $\mathbb{E}[M^*] = 1$, we estimate the pricing errors for each decile portfolio p as follows:

$$\hat{\alpha}_p = \widehat{\mathbb{E}}[xR_p] + \widehat{\mathrm{Cov}}(M^*, xR_p).$$

Table 2: Pricing Errors Under the Three Estimated SDFs

Portfolio	BE/ME SDF	Momentum SDF	Combined SDF
BE/ME D1	0.000005	-0.001505	0.000005
BE/ME D2	0.000006	-0.000095	0.000006
BE/ME D3	0.000006	-0.000217	0.000006
BE/ME D4	0.000006	-0.000772	0.000006
BE/ME D5	0.000006	0.000509	0.000006
BE/ME D6	0.000007	0.001267	0.000007
BE/ME D7	0.000006	0.000287	0.000006
BE/ME D8	0.000008	0.002382	0.000008
BE/ME D9	0.000009	0.003456	0.000009
BE/ME D10	0.000009	0.003204	0.000009
Momentum D1	-0.010147	0.000001	0.000001
Momentum D2	-0.005043	0.000004	0.000004
Momentum D3	-0.003265	0.000004	0.000004
Momentum D4	-0.001482	0.000005	0.000005
Momentum D5	-0.000875	0.000006	0.000006
Momentum D6	-0.000257	0.000006	0.000006
Momentum D7	0.000515	0.000006	0.000006
Momentum D8	0.001908	0.000007	0.000007
Momentum D9	0.002246	0.000008	0.000008
Momentum D10	0.005424	0.000010	0.000010

We observe that the SDF constructed using the BE/ME portfolios prices the BE/ME portfolios well with small pricing errors, but not the MOM portfolios, and vice versa. The SDF based on the combined 20 portfolios prices all 20 portfolios well, which is consistent with the notion that the selection of the asset universe matters in the pricing power of the SDF.

2 (e)

Question 3

3 (a)

3 (b)

3 (c)

3 (d)

3 (e)

Question 4

4 (a)

4 (b)

4 (c)

4 (d)

4 (e)